How Long, and How Sure?

Week 7 — Survival Analysis, Multiple Testing & Wrap-Up (ISLP Ch. 11, Ch. 13)

Ho-min Park & Hyun-Hwan Jeong

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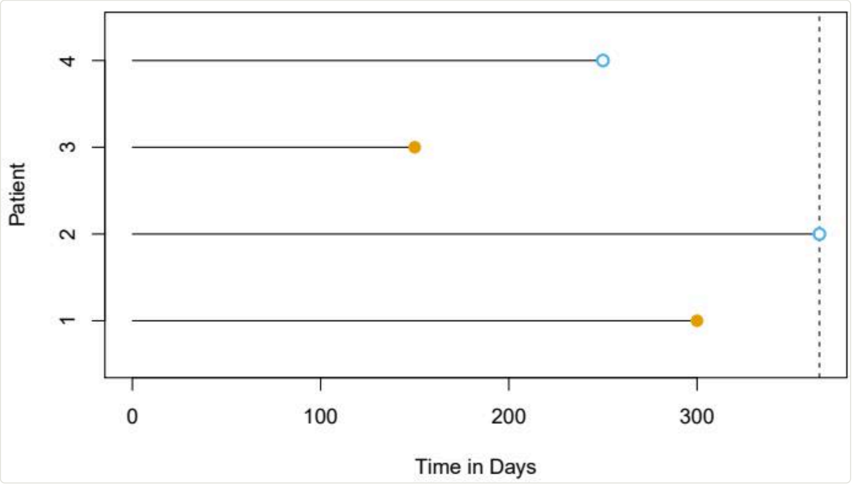
The Finale

Where we are

- **Six weeks in**: regression, classification, validation, regularization, splines, trees, SVMs, neural networks, PCA, clustering
- **Today** — two topics that look niche until the day they aren't:
  - **Ch. 11** — the response is a *waiting time*, and for many subjects **the clock hasn't run out yet**
  - **Ch. 13** — you ran *thousands* of hypothesis tests; how many of your "discoveries" are noise?
- Then: a course wrap-up, and where to go from here

Survival Analysis

The problem: incomplete clocks



Time until death, churn, machine failure, paper publication… For patients 1 and 3 we saw the event. Patient 2 was fine when the study ended; patient 4 dropped out. Their times are **censored** — **we only know the event happens *after* some time**. Throwing them away wastes data *and* biases results toward early events.

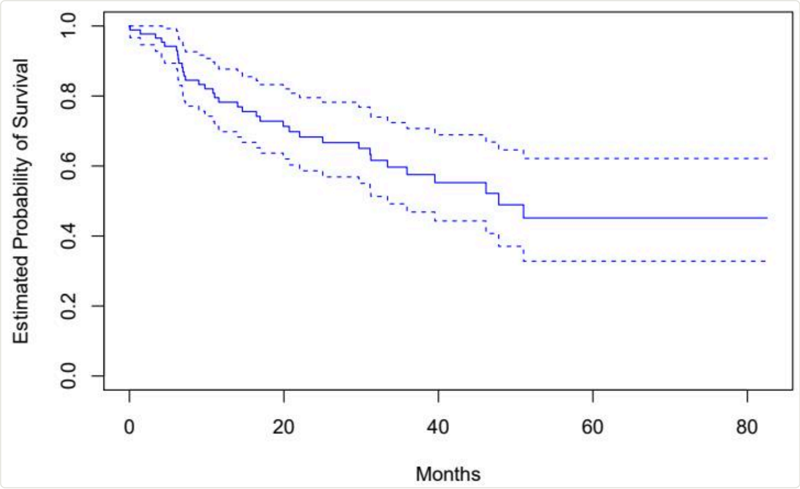
Why not just regress or classify?

- Regression of T on X? The censored observations have no T to regress on — and they're often the *majority*
- "Survived past 5 years, yes/no"? Discards everyone censored before year 5 and all timing detail
- Survival analysis keeps everyone, **each observation contributing exactly what was seen**: an event time y_i and a status δ_i (event vs censored)

Standing assumption: censoring is unrelated to prognosis (*independent censoring*) — worth interrogating in any real study.

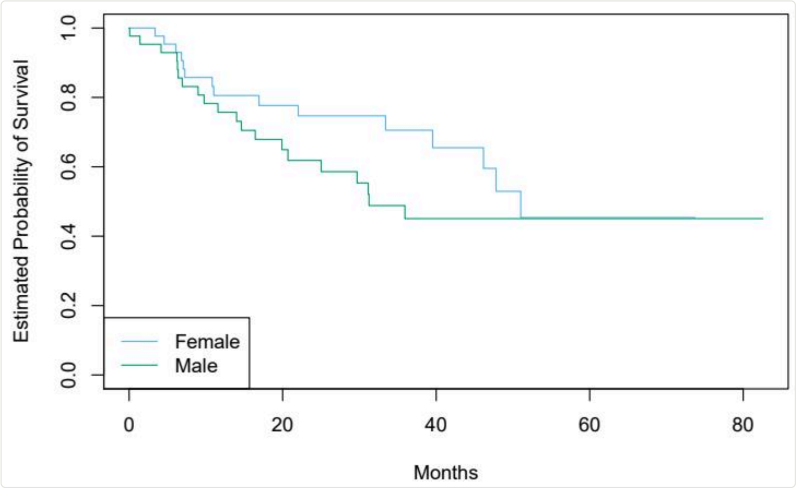
The Kaplan–Meier estimator

$$S(t) = \Pr(T > t) \qquad\qquad \hat{S}(t) = \prod_{j: d_j \le t} \left(1 - \frac{q_j}{r_j}\right)$$



At each observed event time d_j: of the r_j still **at risk**, q_j had the event. Multiply the survival fractions — censored patients count while at risk, then exit gracefully. The result is the staircase everyone in medicine has seen (`BrainCancer` data above).

Comparing groups



Female vs male survival on `BrainCancer` — the curves separate a little. Real or luck? The **log-rank test** compares, at every event time, observed vs expected events per group (expected under H_0: identical survival). Here it finds **no significant difference**.

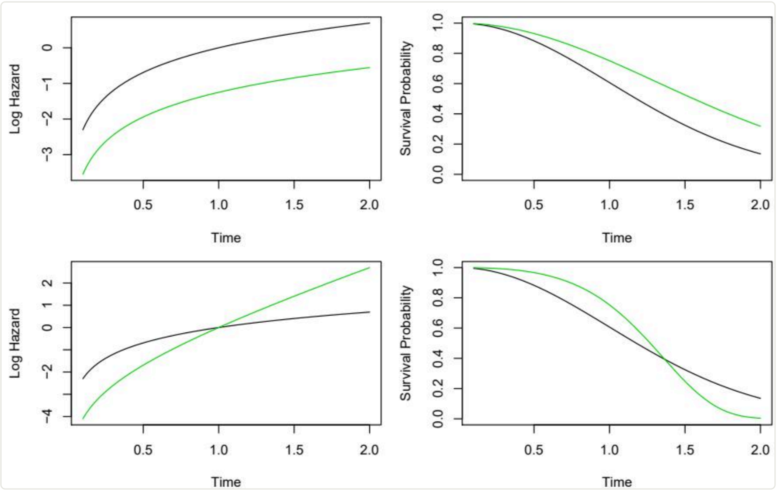
The hazard: risk, right now

$$h(t) = \lim_{\Delta t \to 0} \frac{\Pr(t < T \le t + \Delta t \mid T > t)}{\Delta t}$$

- Read: among those **still alive at t**, the instantaneous event rate
- The hazard and survival function carry the same information in different clothes — high hazard ↔ steeply dropping survival
- Why bother? Because hazards are the natural home for **regression with covariates**…

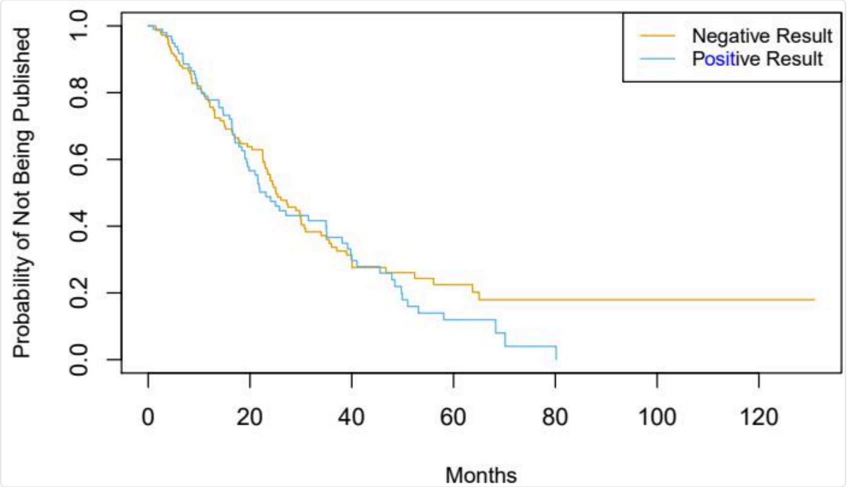
Cox proportional hazards

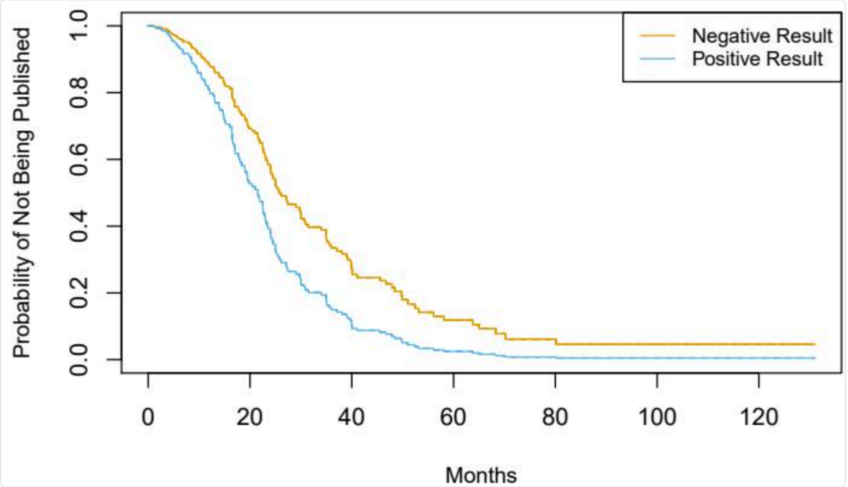
$$h(t \mid x_i) = h_0(t) \exp\left(\sum_{j=1}^{p} x_{ij}\beta_j\right)$$



A baseline hazard $h_0(t)$ — never specified! — scaled up or down by the covariates: one unit more x_j multiplies the hazard by e^{β_j}, **at all times equally** (top row). Estimated by *partial likelihood*, which sidesteps h_0 entirely. Bottom row: what a violation looks like — crossing curves.

A story in two figures: the Publication data





Raw KM curves: positive vs negative trial results reach publication at **nearly the same rate**…

…but adjust for the other covariates in a Cox model and the gap is **dramatic**: positive results publish far sooner.

The toolbox extends as you'd hope

- **Lasso-penalized Cox** — high-dimensional survival ($p \gg n$ genomics), λ by cross-validation: Week 3 machinery, survival edition
- **Survival trees & random survival forests** — Week 4, survival edition
- **Time-dependent covariates** — the Cox partial likelihood handles them naturally
- Checking proportional hazards: plot stratified log hazards; if curves cross, stratify or extend the model

This week's lab, part 1

```
from lifelines import KaplanMeierFitter, CoxPHFitter
from lifelines.statistics import logrank_test

km = KaplanMeierFitter().fit(df['time'], df['status'])
km.plot_survival_function()

cox = CoxPHFitter().fit(df, 'time', 'status')
cox.print_summary()   # hazard ratios = exp(coef)
```

Lab 11.8: `BrainCancer` KM curves and log-rank test, multivariate Cox, and the `Publication` story you just saw.

Multiple Testing

One hypothesis test, honestly stated

1. Set H_0 (e.g. $\mu_1 = \mu_2$: "no difference") and H_a
2. Compute a test statistic T
3. The **p-value**: the probability of a T **at least this extreme, *if* H_0 were true** — not the probability H_0 is true!
4. Reject if p is below your threshold α

	H_0 **true**	H_0 **false**
Reject	Type I error (α)	correct — "power"
Don't reject	correct	Type II error

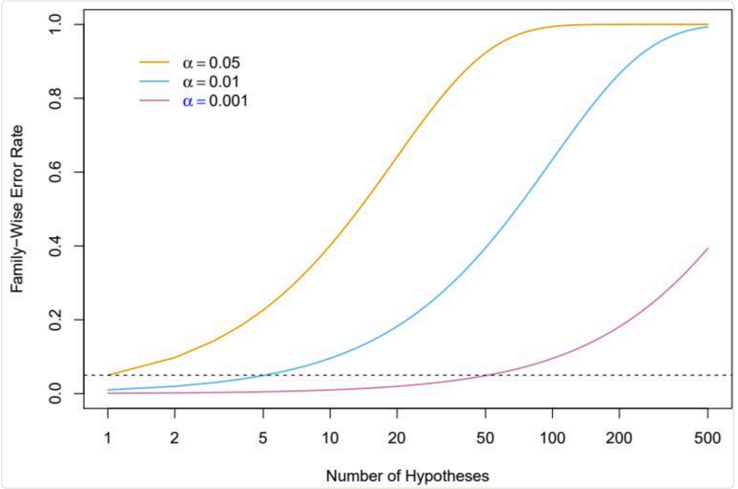
Test enough hypotheses and noise looks like signal

At $\alpha = 0.05$, each true null has a 5% false-alarm rate. Test m independent true nulls and *something* rejects with probability $1 - 0.95^m$:

- $m = 10$: 40% chance of a false discovery
- $m = 100$: 99.4%
- Flip 1,024 fair coins ten times each — **expect one to look "biased"**, guaranteed

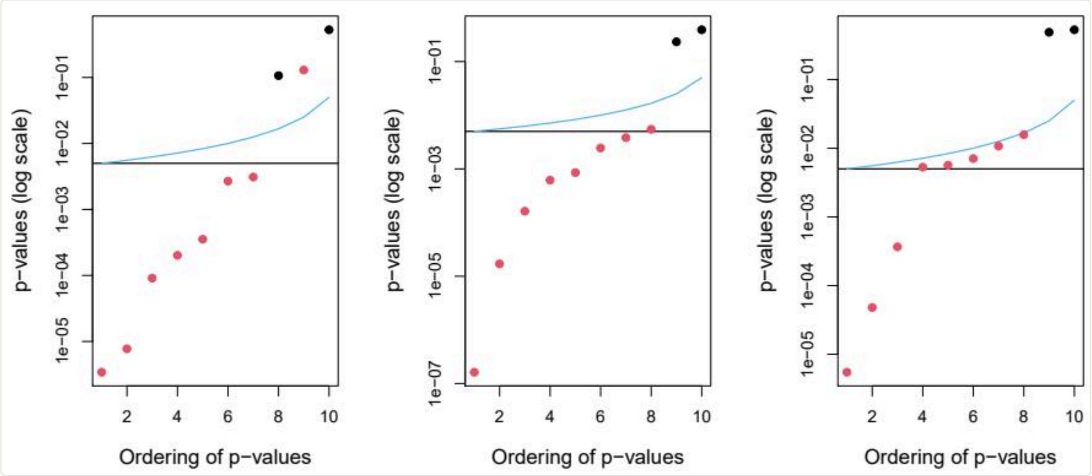
Modern data analysis *is* multiple testing: every gene, every A/B variant, every subgroup, every model comparison.

The family-wise error rate



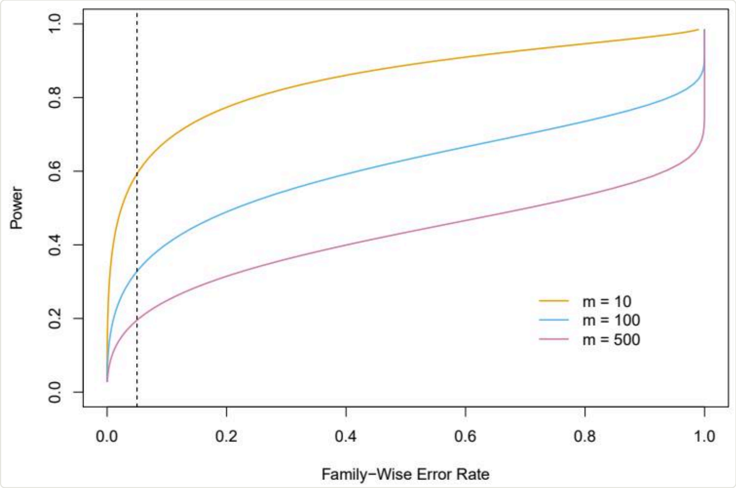
FWER = probability of making **even one** Type I error among the m tests. At per-test $\alpha = 0.05$ it races to 1 as m grows (orange). To keep FWER at 0.05 with $m = 50$ tests, each test must clear $\alpha = 0.001$ (purple).

Controlling FWER: Bonferroni and Holm



Bonferroni: reject only if $p_j \leq \alpha/m$ — simple, assumption-free, brutal. **Holm** steps down through the sorted p-values, relaxing the bar as it goes (blue vs black line) — **always rejects at least as much**, same FWER guarantee. No reason to prefer Bonferroni, ever.

The price of certainty



Power at FWER = 0.05 collapses as m grows: with 500 tests you'll catch only a small fraction of the *real* effects.

"Never a single false positive" is **the wrong goal when screening thousands of candidates** — which motivates a different target…

The false discovery rate

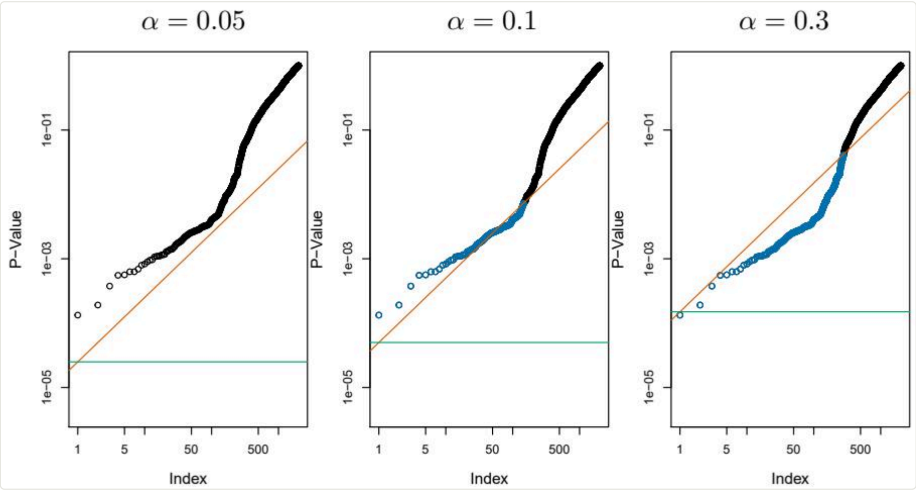
Reframe: among the hypotheses I *reject* — my claimed discoveries — **what fraction are false?**

$$\mathrm{FDR} = E\left[\frac{\text{false rejections}}{\text{total rejections}}\right]$$

FWER asks "any mistakes at all?"; FDR asks "is my discovery list mostly real?" Accepting, say, 10% contamination buys enormous power — the standard bargain in genomics and A/B testing.

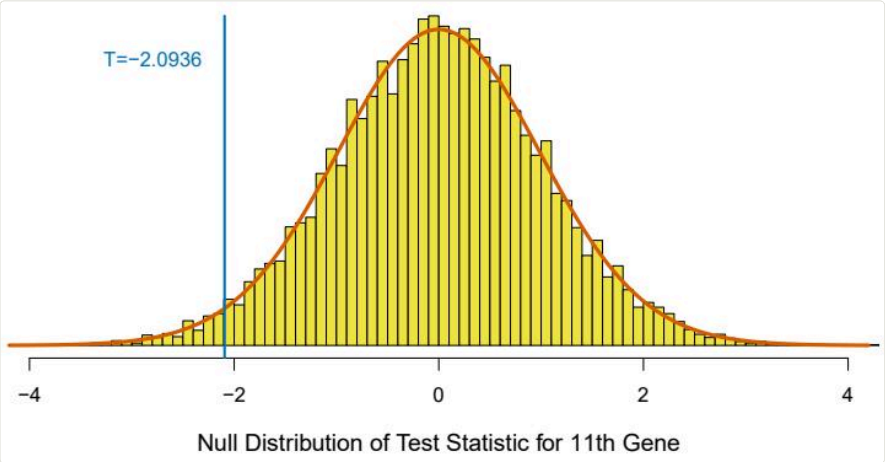
Benjamini–Hochberg: an FDR in three lines

Sort the p-values; reject $H_{(j)}$ for all j up to the largest j with $p_{(j)} < qj/m$.



2,000 fund managers: Bonferroni-style FWER control (green line) rejects almost nothing; **BH** at $q = 0.1$ (orange) declares **146 discoveries** — of which we expect at most ~15 to be false. That's the trade in one picture.

When theory fails: resample the null



The p-value needs a null distribution. Small samples, weird statistics? — **permute** the group labels many times and recompute T: the histogram *is* the null distribution (here on the `Khan` gene data, nearly matching theory). Week 3's resampling spirit, **one last encore**.

This week's lab, part 2

```
from statsmodels.stats.multitest import multipletests

reject_holm, p_holm, *_ = multipletests(pvals, alpha=0.05,
                                        method='holm')
reject_bh, q_vals, *_ = multipletests(pvals, alpha=0.1,
                                      method='fdr_bh')
```

Lab 13.6: FWER vs FDR on the `Fund` data, then permutation p-values on `Khan` — count your discoveries under each regime.

Course Wrap-Up

Seven weeks, one idea

Everything we did was the same move in different costumes: **choose a family of models, fit it to training data, and honestly estimate how it does on data it hasn't seen**.

- The **bias-variance trade-off** was the physics
- **Cross-validation** was the measuring device
- **Regularization** (λ, pruning, dropout, margins…) was the brake
- Everything else — splines, trees, kernels, layers — was a hypothesis about the *shape* of f

What you can now do

Week	You learned to…
1	frame prediction vs inference; respect test error
2	fit & read linear and logistic models, LDA, confusion matrices
3	cross-validate anything; bootstrap anything; ridge & lasso
4	bend fits with splines & GAMs; grow, prune, and ensemble trees
5	tune SVMs with kernels; build and read neural networks & CNNs
6	model sequences; fit networks honestly; PCA & clustering
7	handle censored time-to-event data; survive your own p-values

Where to go from here

- **Depth**: *The Elements of Statistical Learning* (Hastie, Tibshirani, Friedman) — ISLP's rigorous older sibling, free online
- **Breadth**: causal inference (why regression ≠ causation, formally), Bayesian methods, time series
- **Practice**: enter a Kaggle competition with *linear regression first*; add complexity only when CV says so
- **Habit**: for every model you ever ship, be able to answer **"what's your test error, and how do you know?"**

Thank you

Seven Fridays, thirteen chapters, one toolkit.

It's been a pleasure — keep the repo, keep the labs, and keep in touch.

— Ho-min & Hyun-Hwan